

LECTURE NOTES FOR SURV 616

Statistical Methods II

Categorical Data Analysis

CLASS #12

OVERVIEW: In this class we will first discuss modeling categorical data where we clearly define a response variable. We begin by introducing Logit models, and examine their relationship to log-linear models. We then introduce logistic regression models which are like logit models, except they deal with models at the micro level instead of the macro level.

1. Binary Logit Models and Relationships to Log-Linear Models (Agresti, section 6.5)

In log-linear modeling we saw that the intent was to model the response probability vector p_{ijk} , for example, without specifically distinguishing between a response variable and explanatory variables as we do with regression models. In this Class we will examine models which relate a dichotomous response variable to a set of explanatory variables in a way which is similar to what we have done with regression models. We will do this through the *log odds ratio*

$$(1.1) \quad \ln\left(\frac{p}{1-p}\right)$$

where p was the binary response probability. The log odds ratio is modeled by a linear combination of explanatory variables where the unknown parameters in the linear combination are the regression coefficients. We will be particularly interested in estimating the regression coefficients, since we will interpret them as incremental changes in the log odds ratio due to an incremental change in one of the explanatory variables, while keeping all of the other explanatory variables constant.

We begin by discussing logit modeling where we model frequencies and proportions. For the purposes of illustration, assume that we have a 3-way table which is $2 \times K$ where the first variable has two levels, and we want to treat the first variable as the response, or dependent variable, and the other two variables as the predictor, or independent variables. The saturated

log-linear model for such a table is given by

$$(1.2) \quad \ln(m_{ijk}) = \mu + \mu_{1(i)} + \mu_{2(j)} + \mu_{3(k)} + \mu_{12(ij)} + \mu_{13(ik)} + \mu_{23(jk)} + \mu_{123(ijk)}$$

Since the response variable has only two levels we can write

$$(1.3) \quad p_{2|jk} = 1 - p_{1|jk}$$

so the log odds ratio for the response probability $p_{1|jk}$ is given by

$$(1.4) \quad \ln\left(\frac{p_{1|jk}}{1 - p_{1|jk}}\right) = \ln\left(\frac{p_{1|jk}}{p_{2|jk}}\right)$$

where we are thinking of modeling the response probability for each combination of the $J \times K$ interactions. But note that

$$(1.5) \quad \ln\left(\frac{p_{1|jk}}{1 - p_{1|jk}}\right) = \ln\left(\frac{m_{1|jk}}{m_{2|jk}}\right) = \ln(m_{1|jk}) - \ln(m_{2|jk})$$

where we assumed that $m_{ijk} = np_{ijk}$ under multinomial sampling. Therefore, in terms of the log-linear model in (1.2), we can write

$$(1.6)$$

$$\ln\left(\frac{p_{1|jk}}{1 - p_{1|jk}}\right) = \left[\mu + \mu_{1(1)} + \mu_{2(j)} + \mu_{3(k)} + \mu_{12(1j)} + \mu_{13(1k)} + \mu_{23(jk)} + \mu_{123(ijk)} + \mu_{1(i)} + \mu_{123(1jk)} \right] \\ - \left[\mu + \mu_{1(2)} + \mu_{2(j)} + \mu_{3(k)} + \mu_{12(2j)} + \mu_{13(2k)} + \mu_{23(jk)} + \mu_{123(ijk)} + \mu_{1(i)} + \mu_{123(2jk)} \right]$$

so that

$$\begin{aligned}
(1.7) \quad \ln \left(\frac{p_{ijk}}{1 - p_{ijk}} \right) &= w + w_{2(j)} + w_{3(k)} + w_{23(jk)} \\
w &= (\mu_{1(1)} - \mu_{1(2)}) \\
w_{2(j)} &= (\mu_{12(1j)} - \mu_{12(2j)}) \\
w_{3(k)} &= (\mu_{13(1k)} - \mu_{13(2k)}) \\
w_{23(jk)} &= (\mu_{123(1jk)} - \mu_{123(2jk)})
\end{aligned}$$

This is the form of a general logit model, with w -terms corresponding to main effects and interactions for the independent variables. Note that the first order term $\mu_{23(jk)}$ does not appear in the logit model, since it fell out by subtraction. In general, logit modeling ignores the relationships between the independent variables, while log-linear models explicitly model relationships between the independent variables.

In general we can associate a log-linear model to a logit model. For example

Complete Interaction Logit Model
Corresponds to a Saturated Log-linear Model

$$\begin{aligned}
(1.8) \quad \ln \left(\frac{p_{ijk}}{1 - p_{ijk}} \right) &= w + w_{2(j)} + w_{3(k)} + w_{23(jk)} \\
\ln(m_{ijk}) &= \mu + \mu_{1(i)} + \mu_{2(j)} + \mu_{3(k)} + \mu_{12(ij)} + \mu_{13(ik)} + \mu_{23(jk)} + \mu_{123(ijk)}
\end{aligned}$$

Two Main Effects Logit Model
Corresponds to a No Second Order Log-linear Model

$$\begin{aligned}
(1.9) \quad \ln \left(\frac{p_{ijk}}{1 - p_{ijk}} \right) &= w + w_{2(j)} + w_{3(k)} \\
\ln(m_{ijk}) &= \mu + \mu_{1(i)} + \mu_{2(j)} + \mu_{3(k)} + \mu_{12(ij)} + \mu_{13(ik)} + \mu_{23(jk)}
\end{aligned}$$

One Main Effect Logit Model
Corresponds to a Conditional Independence Log-linear Model

$$(1.10) \quad \ln \left(\frac{p_{1jk}}{1 - p_{1jk}} \right) = w + w_{2(j)}$$

$$\ln(m_{ijk}) = \mu + \mu_{1(i)} + \mu_{2(j)} + \mu_{3(k)} + \mu_{12(ij)} + \mu_{23(jk)}$$

The (1,3) term is missing; $u_{2(j)}$, $u_{3(k)}$, $u_{23(jk)}$ subtracted out in (1.6).

Overall Effect (Only) Logit Model
Corresponds to a Pairwise Independence Model

$$(1.11) \quad \ln \left(\frac{p_{1jk}}{1 - p_{1jk}} \right) = w$$

$$\ln(m_{ijk}) = \mu + \mu_{1(i)} + \mu_{2(j)} + \mu_{3(k)} + \mu_{23(jk)}$$

In general the G^2 goodness of fit statistic corresponding to each of the logit models above is equal to the G^2 statistic for the associated log-linear model.

We have been discussing logit models for 3-Way tables, but the same kind of description holds for higher-way tables as well. We next present an example to illustrate some of the ideas of logit modeling.

Example 1. We will again examine the detergent preference data described in the previous notes. A sample of 1008 consumers was questioned about their preference of one of two types of detergents, Brand X and Brand M in a blind trial. The responses are cross classified by the following variables

- (W) Water Softness : (1) Soft, (2) Medium, (3), Hard
 (B) Brand Preference : (1) Brand X, (2) Brand M
 (P) Previous Experience : (1) User of Brand M, (2) Nonuser of Brand M
 (T) Temperature : (1) High, (2) Low

So, we have a 3 × 2 × 2 × 2 table. The data are given below

Water Softness	Brand Preference	Previous User of Brand M		Previous Nonuser of Brand M	
		High Temperature	Low Temperature	High Temperature	Low Temperature
Soft	X	19	57	29	63
	M	29	49	27	53
Medium	X	23	47	33	66
	M	47	55	23	50
Hard	X	24	37	42	68
	M	43	52	30	42

In this example it makes sense to consider brand preference as dependent variable, and the other three variables as independent variables. In addition we will assume that the marginal totals n_{i+kl} are fixed by design, so we will consider log-linear models which include the term

λ_{ikl}^{WPT} in the model, but not the term λ_{ijkl}^{WBPT} .

For the purpose of illustration we will consider fitting two models

Model 1

$$\ln\left(\frac{p_{ijkl}}{1 - p_{ijkl}}\right) = w + w_{1i} + w_{3(k)} + w_{4(l)} + w_{13(ik)} + w_{14(il)} + w_{34(kl)}$$

$$\begin{aligned} \ln(m_{ijkl}) = & \lambda + \lambda_i^W + \lambda_j^B + \lambda_k^P + \lambda_l^T + \lambda_{ij}^{WB} + \lambda_{jl}^{BT} + \lambda_{jk}^{BP} \\ & + \lambda_{il}^{WT} + \lambda_{ik}^{WP} + \lambda_{kl}^{PT} + \lambda_{ijK}^{WBP} + \lambda_{ijl}^{WBT} + \lambda_{jkl}^{BPT} + \lambda_{ikl}^{WPT} \end{aligned}$$

Model 2

$$\ln\left(\frac{p_{ijkl}}{1 - p_{ijkl}}\right) = w + w_{1i} + w_{3(k)} + w_{4(l)}$$

$$\begin{aligned} \ln(m_{ijkl}) = & \lambda + \lambda_i^W + \lambda_j^B + \lambda_k^P + \lambda_l^T + \lambda_{ij}^{WB} + \lambda_{jl}^{BT} + \lambda_{jk}^{BP} \\ & + \lambda_{il}^{WT} + \lambda_{ik}^{WP} + \lambda_{kl}^{PT} + \lambda_{ikl}^{WPT} \end{aligned}$$

where we have written the log-linear models in the λ -term notation. The SAS code below shows how to fit both models in terms of their log-linear representation and their logit model

representation.

```
data set1; input W B P T
  wt;cards;
  1 1 1 1 19
  1 1 1 2 57
  1 1 2 1 29
  1 1 2 2 63
  1 2 1 1 29
  1 2 1 2 49
  1 2 2 1 27
  1 2 2 2 53
  2 1 1 1 23
  2 1 1 2 47
  2 1 2 1 33
  2 1 2 2 66
  2 2 1 1 47
  2 2 1 2 55
  2 2 2 1 23
  2 2 2 2 50
  3 1 1 1 24
  3 1 1 2 37
  3 1 2 1 42
  3 1 2 2 68
  3 2 1 1 43
  3 2 1 2 52
  3 2 2 1 30
  3 2 2 2 42
  ;
proc catmod;
  title 'Log-Linear Analysis for Model 1';
  weight wt;
  model B*W*T*P = _response_;
  loglin B W T P B*W B*T B*P W*T W*P T*P B*W*P B*W*T B*P*T W*T*P;

proc catmod;
  title 'Logit Analysis for Model 1';
  weight wt;
  model B = W P T W*P W*T P*T;

run;

proc catmod;
  title 'Log-Linear Analysis for Model 2';
  weight wt;
  model B*W*T*P = _response_;
  loglin B W T P B*W B*T B*P W*T W*P T*P W*T*P;

proc catmod;
  title 'Logit Analysis for Model 2';
  weight wt;
  model B = W P T;

run;
```

Below we give portions of the output from the SAS code.

Log-Linear Analysis for Model 1
The CATMOD Procedure

Maximum Likelihood Analysis of Variance

Source	DF	Chi-Square	Pr > ChiSq
B	1	0.22	0.6389
W	2	0.94	0.6240
T	1	72.78	<.0001
P	1	1.57	0.2096
B*W	2	0.12	0.9423
B*T	1	3.75	0.0527
B*P	1	21.50	<.0001
W*T	2	5.68	0.0584
W*P	2	1.51	0.4697
T*P	1	0.59	0.4415
B*W*P	2	4.57	0.1017
B*W*T	2	0.16	0.9223
B*T*P	1	2.21	0.1368
W*T*P	2	1.37	0.5029
Likelihood Ratio	2	0.74	0.6917

Analysis of Maximum Likelihood Estimates

Parameter	Estimate	Standard Error	Chi-Square	Pr > ChiSq
B 1	-0.0158	0.0336	0.22	0.6389
W 1	-0.0467	0.0481	0.94	0.3319
W 2	0.0251	0.0470	0.29	0.5929
T 1	-0.2863	0.0336	72.78	<.0001
P 1	-0.0422	0.0336	1.57	0.2096
B*W 1 1	0.0128	0.0483	0.07	0.7905
B*W 1 2	-0.0153	0.0472	0.11	0.7452
B*T 1 1	-0.0650	0.0336	3.75	0.0527
B*P 1 1	-0.1553	0.0335	21.50	<.0001
W*T 1 1	-0.0971	0.0479	4.11	0.0427
W*T 2 1	-0.00056	0.0464	0.00	0.9905
W*P 1 1	-0.0232	0.0481	0.23	0.6291
W*P 2 1	0.0574	0.0471	1.48	0.2232
T*P 1 1	0.0259	0.0336	0.59	0.4415
B*W*P 1 1 1	0.0942	0.0456	4.27	0.0387
B*W*P 1 2 1	-0.0246	0.0451	0.30	0.5851
B*W*T 1 1 1	-0.0187	0.0482	0.15	0.6984
B*W*T 1 2 1	0.00506	0.0472	0.01	0.9146
B*T*P 1 1 1	-0.0499	0.0336	2.21	0.1368
W*T*P 1 1 1	-0.0471	0.0482	0.95	0.3290
W*T*P 2 1 1	0.0500	0.0472	1.12	0.2896

Logit Analysis for Model 1

Maximum Likelihood Analysis of Variance

Source	DF	Chi-Square	Pr > ChiSq
Intercept	1	0.22	0.6389
W	2	0.12	0.9423
P	1	21.50	<.0001
T	1	3.75	0.0527
W*P	2	4.57	0.1017
W*T	2	0.16	0.9223
P*T	1	2.21	0.1368
Likelihood Ratio	2	0.74	0.6917

Analysis of Maximum Likelihood Estimates

Parameter		Estimate	Standard Error	Chi-Square	Pr > ChiSq
Intercept		-0.0315	0.0671	0.22	0.6389
W	1	0.0256	0.0965	0.07	0.7905
	2	-0.0307	0.0944	0.11	0.7452
P	1	-0.3106	0.0670	21.50	<.0001
T	1	-0.1301	0.0671	3.75	0.0527
W*P	1 1	0.1884	0.0911	4.27	0.0387
	2 1	-0.0493	0.0903	0.30	0.5851
W*T	1 1	-0.0373	0.0963	0.15	0.6984
	2 1	0.0101	0.0944	0.01	0.9146
P*T	1 1	-0.0999	0.0671	2.21	0.1368

Log-Linear Analysis for Model 2

Maximum Likelihood Analysis of Variance Table

Source	DF	Chi-Square	Pr > ChiSq
B	1	0.21	0.6491
W	2	1.48	0.4781
T	1	71.15	<.0001
P	1	1.32	0.2506
B*W	2	0.22	0.8976
B*T	1	3.73	0.0534

Log-Linear Analysis for Model 2

Maximum Likelihood Analysis of Variance Table

Source	DF	Chi-Square	Pr > ChiSq
B*P	1	19.70	<.0001
W*T	2	6.18	0.0454
W*P	2	1.52	0.4667
T*P	1	0.64	0.4247
W*T*P	2	1.61	0.4461

Likelihood Ratio 7 **8.23** **0.3129**

Analysis of Maximum Likelihood Estimates

Parameter		Estimate	Standard Error	Chi-Square	Pr > ChiSq
B	1	-0.0151	0.0332	0.21	0.6491
W	1	-0.0578	0.0476	1.47	0.2253
	2	0.0263	0.0463	0.32	0.5691
T	1	-0.2787	0.0330	71.15	<.0001
P	1	-0.0382	0.0333	1.32	0.2506
B*W	1 1	0.0153	0.0455	0.11	0.7370
	1 2	-0.0200	0.0449	0.20	0.6554
B*T	1 1	-0.0642	0.0332	3.73	0.0534
B*P	1 1	-0.1418	0.0319	19.70	<.0001
W*T	1 1	-0.1014	0.0477	4.52	0.0336
	2 1	0.000310	0.0464	0.00	0.9947
W*P	1 1	-0.0228	0.0481	0.23	0.6352
	2 1	0.0570	0.0467	1.49	0.2225
T*P	1 1	0.0266	0.0333	0.64	0.4247
W*T*P	1 1 1	-0.0516	0.0476	1.18	0.2783
	2 1 1	0.0523	0.0463	1.28	0.2581

Log-Linear Analysis for Model 2

Maximum Likelihood Analysis of Variance

Source	DF	Chi-Square	Pr > ChiSq
Intercept	1	0.21	0.6491
W	2	0.22	0.8976
P	1	19.70	<.0001
T	1	3.73	0.0534
Likelihood Ratio	7	8.23	0.3129

Logit Analysis for Model 2

The CATMOD Procedure

Analysis of Maximum Likelihood Estimates

Parameter		Estimate	Standard Error	Chi-Square	Pr > ChiSq
Intercept		-0.0302	0.0663	0.21	0.6491
W	1	0.0306	0.0911	0.11	0.7370
	2	-0.0400	0.0897	0.20	0.6553
P	1	-0.2835	0.0639	19.70	<.0001
T	1	-0.1283	0.0664	3.73	0.0534

Note that for both models, that the likelihood ratio statistics (which have been bolded in the

output) are the same for the log-linear representation and the logit representation as they should be. We have also included the estimates for the λ -terms and the w -terms so that we can show the connection between the logit model parameters and the log-linear model parameters. For example, in Model 1 the relationship

$$w = \lambda_1^B - \lambda_2^B$$

but we know that the λ -terms must satisfy the constraint

$$\lambda_1^B + \lambda_2^B = 0$$

which implies

$$\lambda_2^B = -\lambda_1^B$$

Therefore

$$w = \lambda_1^B - \lambda_2^B = \lambda_1^B - (-\lambda_1^B) = 2\lambda_1^B$$

We have bolded the estimates for w and λ_1^B in the output, which are given by

$$\hat{\lambda}_1^B = -0.0158$$

$$\hat{w} = -0.0315$$

which follow the relationship above (within rounding error). From Model 2 it looks like we might be able to reduce down to the model

Model 3

$$\ln\left(\frac{P_{i1kl}}{1 - P_{i1kl}}\right) = w + w_{3(k)}$$

$$\ln(m_{ijkl}) = \lambda + \lambda_i^W + \lambda_j^B + \lambda_k^P + \lambda_l^T + \lambda_{jk}^{BP} + \lambda_{il}^{WT} + \lambda_{ik}^{WP} + \lambda_{kl}^{PT} + \lambda_{ikl}^{WPT}$$

From this we have

$$\ln(m_{i1kl}) = \lambda + \lambda_i^W + \lambda_1^B + \lambda_k^P + \lambda_l^T + \lambda_{1k}^{BP} + \lambda_{il}^{WT} + \lambda_{ik}^{WP} + \lambda_{kl}^{PT} + \lambda_{ikl}^{WPT}$$

$$\ln(m_{i2kl}) = \lambda + \lambda_i^W + \lambda_2^B + \lambda_k^P + \lambda_l^T + \lambda_{2k}^{BP} + \lambda_{il}^{WT} + \lambda_{ik}^{WP} + \lambda_{kl}^{PT} + \lambda_{ikl}^{WPT}$$

$$\Rightarrow \ln\left(\frac{m_{i1kl}}{m_{i2kl}}\right) = (\lambda_1^B - \lambda_2^B) + (\lambda_{1k}^{BP} - \lambda_{2k}^{BP}) \approx w + w_{3(k)}$$

which says that the odds ratio for consumer preference could be modeled entirely as a function of *Previous Experience* alone. We give the SAS code and output from the code below for fitting the logit model and the log-linear form of the model (where we omit repeating the data step which is the same as above)

(SAME DATA STEP AS ABOVE)

```
proc catmod;
  title 'Log-Linear Analysis for Model 3';
  weight wt;
  model B*W*T*P = _response_;
  loglin B W T P B*P W*T W*P T*P W*T*P;

proc catmod;
  title 'Logit Analysis for Model 3';
  weight wt; model B = P;

run;
```

Log-Linear Analysis for Model 3

Maximum Likelihood Analysis of Variance

Source	DF	Chi-Square	Pr > ChiSq
B	1	0.00	0.9565
W	2	1.49	0.4757
T	1	71.00	<.0001
P	1	1.16	0.2808
B*P	1	20.37	<.0001
W*T	2	6.24	0.0441
W*P	2	1.70	0.4274
T*P	1	1.17	0.2796
W*T*P	2	1.61	0.4462
Likelihood Ratio	10	12.24	0.2690

Analysis of Maximum Likelihood Estimates

Parameter		Estimate	Standard Error	Chi-Square	Pr > ChiSq
B	1	0.00174	0.0319	0.00	0.9565
W	1	-0.0580	0.0476	1.48	0.2235
	2	0.0267	0.0463	0.33	0.5634
T	1	-0.2777	0.0330	71.00	<.0001
P	1	-0.0359	0.0333	1.16	0.2808
B*P	1 1	-0.1438	0.0319	20.37	<.0001
W*T	1 1	-0.1024	0.0476	4.62	0.0316
	2 1	0.00157	0.0463	0.00	0.9730
W*P	1 1	-0.0249	0.0476	0.27	0.6005
	2 1	0.0598	0.0463	1.67	0.1964
T*P	1 1	0.0356	0.0330	1.17	0.2796
W*T*P	1 1 1	-0.0516	0.0476	1.18	0.2783
	2 1 1	0.0523	0.0463	1.28	0.2582

Logit Analysis for Model 3

Maximum Likelihood Analysis of Variance

Source	DF	Chi-Square	Pr > ChiSq
Intercept	1	0.00	0.9565
P	1	20.37	<.0001
Likelihood Ratio	0	.	.

Analysis of Maximum Likelihood Estimates

Parameter		Estimate	Standard Error	Chi-Square	Pr > ChiSq
Intercept		0.00348	0.0637	0.00	0.9565
P	1	-0.2875	0.0637	20.37	<.0001

We have bolded the likelihood ratio statistics for the log-linear model and the logit model. Note that now the two statistics are *not* the same. The G^2 statistic for the log-linear model is correct, whereas the likelihood ratio statistic for the logit model reports zero degrees of freedom. The reason for the unusual likelihood ratio statistic for the logit model is that it *thinks* that the data comes from a $2 \times 2 \times 2$ table defined by the two levels for *Brand* and the two levels for *Previous Use* so Model 3 looks like a *saturated model*. Therefore when looking at G^2 statistics, you should refer to the associated log-linear model in all cases to be safe. Interestingly the parameter estimates and standard errors for the logit model and log-linear model are directly comparable. For example, we have

$$w = \lambda_1^B - \lambda_2^B = 2\lambda_1^B$$

$$w_{3(k)} = \lambda_{1k}^{BP} - \lambda_{2k}^{BP} = 2\lambda_{1k}^{BP}$$

because of the summation constraints. We have bolded the estimates (and their standard errors) in the output and they are given by

$$\hat{w} = 0.00348$$

$$\hat{\lambda}_1^B = 0.00174$$

$$\hat{w}_{3(k)} = -0.2875$$

$$\lambda_{1k}^{BP} = -0.1438$$

and the relationship above corresponds, in terms of the estimates, within rounding error. The problem with the likelihood ratio statistic for the logit formulation of the reduced model can be corrected by using the *POPULATION* statement in PROC CATMOD. For example, if we had run the reduced model with the code

```
proc catmod;
  title 'Logit Analysis for Model 3';
  weight wt;
  population W T P;
  model B = P;
run;
```

We would have gotten the *correct* output below.

Logit Analysis for Model 3

Maximum Likelihood Analysis of Variance

Source	DF	Chi-Square	Pr > ChiSq
Intercept	1	0.00	0.9565
P	1	20.37	<.0001
Likelihood Ratio	10	12.24	0.2690

Analysis of Maximum Likelihood Estimates

Parameter	Estimate	Standard Error	Chi-Square	Pr > ChiSq
Intercept	0.00348	0.0637	0.00	0.9565
P 1	-0.2875	0.0637	20.37	<.0001

The *POPULATION* statement tells SAS to use a different table of frequencies than is implied by the *right hand side* of the *MODEL* statement alone. ■

In the next example we reexamine the death penalty data we looked at in the last homework assignment.

Example 2. The respondents were defendants in homicide indictments in 20 Florida counties during 1976-1977, and cross classified by the variables

D Defendant's Race
V Victim's Race
P Death Party

Defendant's race	Victim's race	Death penalty		Total	Percent
		Yes	No		Yes
White	White	19	132	151	12.6%
	Black	0	9	9	0.0%
Black	White	11	52	63	17.5%
	Black	6	97	103	5.8%
Total defendants	White	30	184	214	14.0%
	Black	6	106	112	5.4%
White Black	Total victims	19	141	160	11.9%
		17	149	166	10.2%
Total	Total	36	290	326	11.0%

It is natural to treat the Death Penalty as the dependent variable, and treat the other two variables as independent variables. We will look at three different models.

Model 1

$$\ln\left(\frac{p_{ij1}}{1-p_{ij1}}\right) = w + w_i^D + w_j^V$$
$$\ln(m_{ijk}) = \lambda + \lambda_i^D + \lambda_j^V + \lambda_k^P + \lambda_{ik}^{DP} + \lambda_{ij}^{DV} + \lambda_{jk}^{VP}$$

Model 2

$$\ln\left(\frac{p_{ij1}}{1-p_{ij1}}\right) = w + w_j^V$$
$$\ln(m_{ijk}) = \lambda + \lambda_i^D + \lambda_j^V + \lambda_k^P + \lambda_{ij}^{DV} + \lambda_{jk}^{VP}$$

Model 3

$$\ln\left(\frac{p_{ij1}}{1-p_{ij1}}\right) = w$$
$$\ln(m_{ijk}) = \lambda + \lambda_i^D + \lambda_j^V + \lambda_k^P + \lambda_{ij}^{DV}$$

We give the SAS code and output below.

```
data set1;
  input D V P wt;
  wt = wt + 0.5;
  cards;
    1 1 1 19
    1 1 2 132
    1 2 1 0
    1 2 2 9
    2 1 1 11
    2 1 2 52
    2 2 1 6
    2 2 2 97
  ;
proc catmod;
  title 'Log-Linear Model 1';
  weight wt;
  model D*V*P = _response_;
  loglin P D V P*D P*V D*V;

run;
```

```

proc catmod;
  title 'Logit Model 1';
  weight wt;
  model P = D V;

run;

proc catmod;
  title 'Log-Linear Model 2';
  weight wt; model D*V*P = _response_;
  loglin P D V P*V D*V;

run;

proc catmod;
  title 'Logit Model 2';
  weight wt;
  model P = V;

run;

proc catmod;
  title 'Log-Linear Model 3';
  weight wt;
  model D*V*P = _response_;
  loglin P D V D*V;

run;

proc catmod;
  title 'Logit Model 3';
  weight wt; model P = ;

run;

```

Maximum Likelihood Analysis of Variance

Source	DF	Chi-Square	Pr > ChiSq
P	1	104.55	<.0001
D	1	13.86	0.0002
V	1	33.08	<.0001
D*P	1	0.97	0.3253
V*P	1	5.67	0.0172
D*V	1	78.57	<.0001
Likelihood Ratio	1	0.01	0.9191

Analysis of Maximum Likelihood Estimates

Parameter	Estimate	Standard Error	Chi-Square	Pr > ChiSq
P	-1.1526	0.1127	104.55	<.0001
D	-0.4461	0.1198	13.86	0.0002
V	0.7959	0.1384	33.08	<.0001
D*P	-0.0968	0.0985	0.97	0.3253

V*P	1	1	0.2932	0.1231	5.67	0.0172
D*V	1	1	0.8114	0.0915	78.57	<.0001

Logit Model 1

MAXIMUM-LIKELIHOOD ANALYSIS-OF-VARIANCE TABLE

Source	DF	Chi-Square	Pr > ChiSq
Intercept	1	104.58	<.0001
D	1	0.97	0.3253
V	1	5.67	0.0172
Likelihood Ratio	1	0.01	0.9191

Analysis of Maximum Likelihood Estimates

Parameter	Estimate	Standard Error	Chi-Square	Pr > ChiSq
Intercept	-2.3051	0.2254	104.58	<.0001
D 1	-0.1937	0.1969	0.97	0.3253
V 1	0.5863	0.2462	5.67	0.0172

Log-Linear Model 2

MAXIMUM-LIKELIHOOD ANALYSIS-OF-VARIANCE TABLE

Source	DF	Chi-Square	Pr > ChiSq
P	1	107.29	<.0001
D	1	16.55	<.0001
V	1	33.42	<.0001
V*P	1	4.66	0.0309
D*V	1	78.01	<.0001
Likelihood Ratio	2	0.96	0.6185

Logit Model 2

MAXIMUM-LIKELIHOOD ANALYSIS-OF-VARIANCE TABLE

Source	DF	Chi-Square	Pr > ChiSq
Intercept	1	107.32	<.0001
V	1	4.66	0.0309
Likelihood Ratio	0	.	.

Analysis of Maximum Likelihood Estimates

Parameter	Estimate	Standard Error	Chi-Square	Pr > ChiSq
Intercept	-2.2566	0.2178	107.32	<.0001
V 1	0.4703	0.2178	4.66	0.0309

Log-Linear Model 3

MAXIMUM-LIKELIHOOD ANALYSIS-OF-VARIANCE TABLE

Source	DF	Chi-Square	Pr > ChiSq
P	1	139.82	<.0001
D	1	16.55	<.0001
V	1	37.92	<.0001
D*V	1	78.01	<.0001
Likelihood Ratio	3	6.37	0.0947

Analysis of Maximum Likelihood Estimates

Parameter		Estimate	Standard Error	Chi-Square	Pr > ChiSq
P	1	-1.0196	0.0862	139.82	<.0001
D	1	-0.3692	0.0908	16.55	<.0001
V	1	0.5589	0.0908	37.92	<.0001
D*V	1 1	0.8017	0.0908	78.01	<.0001

Log-Linear Model 3

Maximum Likelihood Analysis of Variance

Source	DF	Chi-Square	Pr > ChiSq
Intercept	1	139.82	<.0001
Likelihood Ratio	0	.	.

ANALYSIS OF MAXIMUM-LIKELIHOOD ESTIMATES

Parameter	Estimate	Error	Square	Pr > ChiSq
Intercept	-2.0392	0.1725	139.82	<.0001

We get the following G^2 statistics

$$G^2_{Model1} = 0.01 \quad df = 1 \quad p\text{-value} = 0.9191$$

$$G^2_{Model2} = 0.96 \quad df = 2 \quad p\text{-value} = 0.6185$$

$$G^2_{Model3} = 6.37 \quad df = 3 \quad p\text{-value} = 0.0947$$

We can clearly reduce to Model 2 which says that the odds of getting the death penalty depends on the victim's race but not the defendant's race. In particular we have bolded the estimate $w_1^v = 0.4703$ in the output, which indicates that the defendant is more likely to get the death penalty if the victim is white, and less likely if the victim is black. The models do not pick up any $D*V$ interaction. If we use a significance level of less than $\alpha = 0.10$ we can reduce all the way to Model 3 which says that the odds of getting the death penalty does not depend on the

victim's race either. The conclusions from the data set are unclear, in that there would be some people who would go with Model 2 and others would go with Model 3.

■

As another example of how to use the POPULATION statement and logit modeling, we show how to analyze the golf problem from a previous homework example.

Example 3. The homework problem specified a sample of scores from a round of 18 holes for 60 professional golfers who played the same course in the first round of a "hypothetical tournament". The counts below are the number of players who scored better than par on a given hole.

Hole	Par	Better than Par
1	4	13
2	4	14
3	3	5
4	4	15
5	4	11
6	4	13
7	3	2
8	5	20
9	4	11
10	4	14
11	4	10
12	5	17
13	4	13
14	3	8
15	4	10
16	4	15
17	4	12
18	4	9

Our goal was to test the hypothesis that we can model the probabilities of “Better than Par” for each hole as

$$p_i = \pi_3 \text{ for } i = 3, 7, 14$$

$$p_i = \pi_4 \text{ for } i = 1, 2, 4, 5, 6, 9, 10, 11, 13, 15, 16, 17, 18$$

$$p_i = \pi_5 \text{ for } i = 8, 12$$

We can solve the problem in SAS with the following code

```
data set1;
  input H P S wt;
  cards;
    1 4 1 13
    1 4 2 47
    2 4 1 14
    2 4 2 46
    3 3 1 5
    3 3 2 55
    4 4 1 15
    4 4 2 45
    5 4 1 11
    5 4 2 49
    6 4 1 13
    6 4 2 47
    7 3 1 2
    7 3 2 58
    8 5 1 20
    8 5 2 40
    9 4 1 11
    9 4 2 49
    10 4 1 14
    10 4 2 46
    11 4 1 10
    11 4 2 50
    12 5 1 17
    12 5 2 43
    13 4 1 13
    13 4 2 47
    14 3 1 8
    14 3 2 52
    15 4 1 10
    15 4 2 50
    16 4 1 15
    16 4 2 45
    17 4 1 12
    17 4 2 48
    18 4 1 9
    18 4 2 51
  ;
proc catmod;
  weight wt;
  population H;
  model S = P;

run;
```

We get the following output

The CATMOD Procedure

Data Summary

Response	S	Response Levels	2
Weight Variable	wt	Populations	18
Data Set	SET1	Total Frequency	1080
Frequency Missing	0	Observations	36

Population Profiles

Sample	H	Sample Size
1	1	60
2	2	60
3	3	60
4	4	60
5	5	60
6	6	60
7	7	60
8	8	60
9	9	60
10	10	60
11	11	60
12	12	60
13	13	60
14	14	60
15	15	60
16	16	60
17	17	60
18	18	60

Response Profiles

Response	S
1	1
2	2

Maximum Likelihood Analysis of Variance Table

Source	DF	Chi-Square	Pr > ChiSq
Intercept	1	173.80	<.0001
P	2	22.61	<.0001
Likelihood Ratio	15	9.38	0.8570

Analysis of Maximum Likelihood Estimates

Parameter	Estimate	Standard Error	Chi-Square	Pr > ChiSq
-----------	----------	----------------	------------	------------

Intercept		-1.5201	0.1153	173.80	<.0001
P	3	-0.8778	0.1937	20.53	<.0001
	4	0.1656	0.1262	1.72	0.1894

Note the part of the output which gives the *POPULATION PROFILE*, in which SAS correctly recognizes that we have 18 holes and in the *RESPONSE PROFILE* it sees that we have two responses for each hole, and we will model the response probability by a logit model. When we fit a logit model for which the log odds only depends on the *par value* of the hole, we get the correct G^2 statistic with 15 degrees of freedom. The same result as we got by hand in the homework before. .

In the examples above we have concentrated on cases where the independent variables were categorical. In practice we often want to fit logit models where some of the independent variables are continuous. In the next example we illustrate how to fit an *analysis of covariance form* of the logit model.

Example 4. The data in the following table give the numbers of litters of mice with no deaths and at least one death in each of ten categories formed by the combinations of two factors: litter size and type of treatment. The object of the experiment is to determine how treatments *A* and *B* affect the odds of having all mice surviving in a litter (no death). The litter size was recorded because it was expected that the odds of no deaths would decline as the litter size increased.

Data for Example 4			
Number of Deaths			
Litter Size	Treatment	None	At least One
7	A	58	16
	B	75	26
8	A	49	24
	B	58	25
9	A	33	33
	B	45	32
10	A	15	28
	B	39	40
11	A	4	29
	B	5	23

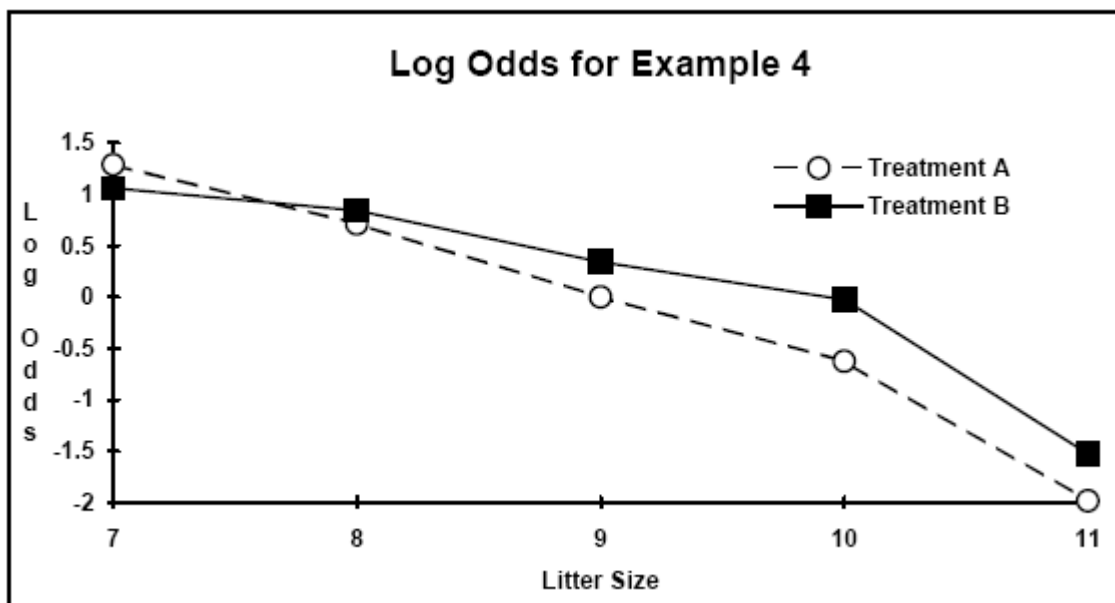
We will denote the variables as

D Number of Deaths
T Treatment
L Litter Size

The first thing to do is graph

$$\ln \left(\frac{n_{1jk}}{n_{2jk}} \right) = \ln \left(\frac{n_{1jk}/n_{+jk}}{n_{2jk}/n_{+jk}} \right) \text{ for } j=1,2 \quad k=1,2,3,4,5$$

where n_{1jk} is the number of cases where *no deaths* occurred for the j -th treatment and k -th litter size.



It appears that the probability of *no deaths* decreases as the litter size increases, and there appears to be an indication that treatment B results in a larger probability of *no deaths*. To investigate this more formally we fit the model

$$\ln \left(\frac{n_{1jk}}{n_{2jk}} \right) = \alpha_j + \beta_j X_k$$

where

$$X_1 = 7, X_2 = 8, X_3 = 9, X_4 = 10, X_5 = 11$$

are the actual litter sizes. We can fit this model with the following SAS code, and the output follows.

```
data set1;
  input L T D wt;
  wt = wt + 0.5;
  cards;
  7 1 1 58
  7 1 2 16
  7 2 1 75
  7 2 2 26
  8 1 1 49
  8 1 2 24
  8 2 1 58
  8 2 2 25
  9 1 1 33
  9 1 2 33
  9 2 1 45
  9 2 2 32
  10 1 1 15
  10 1 2 28
  10 2 1 39
  10 2 2 40
  11 1 1 4
  11 1 2 29
  11 2 1 5
  11 2 2 23
  ;
proc catmod;
  weight wt;
  direct L;
  model D = T L T*L;
run;
```

The CATMOD Procedure

Data Summary

Response	D	Response Levels	2
Weight Variable	wt	Populations	10
Data Set	SET1	Total Frequency	667
Frequency Missing	0	Observations	20

Population Profiles

Sample	T	L	Sample Size
1	1	7	75
2	1	8	74
3	1	9	67
4	1	10	44
5	1	11	34
6	2	7	102
7	2	8	84
8	2	9	78
9	2	10	80
10	2	11	29

Response Profiles

Response	D
1	1
2	2

Maximum Likelihood Analysis

Maximum likelihood computations converged.

Maximum Likelihood Analysis of Variance

Source	DF	Chi-Square	Pr > ChiSq
Intercept	1	81.28	<.0001
T	1	2.27	0.1315
L	1	74.35	<.0001
L*T	1	2.95	0.0857
Likelihood Ratio	6	5.66	0.4628

Analysis of Maximum Likelihood Estimates

Parameter	Estimate	Standard Error	Chi-Square	Pr > ChiSq
Intercept	5.4620	0.6058	81.28	<.0001
T 1	0.9137	0.6058	2.27	0.1315
L	-0.5957	0.0691	74.35	<.0001
L*T 1	-0.1187	0.0691	2.95	0.0857

It appears the coefficient on X_k may not vary significantly by treatment because $L*T$ is not significant, which leads us to consider the *analysis of covariance* logit model

$$\ln\left(\frac{p_{1jk}}{1-p_{1jk}}\right) = \alpha_j + \beta X_K$$

We fit this model with the following SAS code, and the output follows

```
proc catmod;
  weight wt;
  direct L;
  model D = T L;

run;
```

Maximum Likelihood Analysis of Variance

Source	DF	Chi-Square	Pr > ChiSq
Intercept	1	80.52	<.0001
T	1	1.97	0.1609
L	1	73.67	<.0001
Likelihood Ratio	7	8.66	0.2781

Analysis of Maximum Likelihood Estimates

Parameter	Estimate	Standard Error	Chi-Square	Pr > ChiSq
Intercept	5.3142	0.5922	80.52	<.0001
T	-0.1182	0.0843	1.97	0.1609
L	-0.5780	0.0673	73.67	<.0001

We easily accept the hypothesis of the single slope analysis of covariance model, since the likelihood ratio statistic has a *p-value* of 0.2781. No we can test for the presence of a treatment effect beyond that accounted for by the litter size. The bolded lines for *treatment* in the output indicate that there is not a significant treatment effect. To test the model

$$\ln\left(\frac{p_{1jk}}{1-p_{1jk}}\right) = \alpha + \beta X_K$$

we used the following SAS code, with output following. Note that we used the POPULATION statement.

```
proc catmod;
  weight wt;
  direct L;
  population T L;
  model D = L;

run;
```

Maximum Likelihood Analysis of Variance

Source	DF	Chi-Square	Pr > ChiSq
Intercept	1	80.91	<.0001
L	1	73.72	<.0001
Likelihood Ratio	8	10.63	0.2238

Analysis of Maximum Likelihood Estimates

Parameter	Estimate	Standard Error	Chi-Square	Pr > ChiSq
Intercept	5.3108	0.5904	80.91	<.0001
L	-0.5761	0.0671	73.72	<.0001

Notice that the negative on the L estimate agrees with the slope in the figure on p. 23. The likelihood ratio statistic for the model

$$\ln\left(\frac{p_{1jk}}{1-p_{1jk}}\right) = \alpha + \beta X_K$$

of $G^2 = 10.63$, with $p\text{-value} = 0.2283$ indicates that there is no difference in the treatments after taking litter size into account.

2. Logistic Regression Models.

With the usual regression models we often assume the following setup

$$(2.1) \quad \begin{aligned} &(Y_1, \mathbf{X}_1), \dots, (Y_n, \mathbf{X}_n) \text{ independent sample} \\ &(Y_i | \mathbf{X}_i) \sim N(\mu_i, \sigma^2) \\ &\mu_i = \mathbf{X}_i' \beta \end{aligned}$$

where the response variable Y_i is a continuous variable following a Normal distribution, conditional on the p predictors $\mathbf{X}_i$. We often estimate the regression coefficients β by ordinary least squares, that is, the values which minimize the sum of squared deviations

$$(2.2) \quad \sum_{i=1}^n (Y_i - \mathbf{X}_i' \beta)^2$$

We justified this from a heuristic point of view, but we also justified it more formally based upon the *method of maximum likelihood*. The likelihood, or probability density function for the sample as a function of the parameters, for the response conditional upon the predictors as given in (2.1) can be written

$$(2.3) \quad L(\boldsymbol{\beta}, \sigma^2) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left\{ \frac{(Y_i - \mathbf{X}_i' \boldsymbol{\beta})^2}{-2\sigma^2} \right\}$$

or more conveniently, the log-likelihood can be written

$$(2.4) \quad \log L(\boldsymbol{\beta}, \sigma^2) = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log(\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (Y_i - \mathbf{X}_i' \boldsymbol{\beta})^2$$

Maximizing the likelihood, is equivalent to maximizing the log-likelihood since the *log* function is a one-to-one function. Notice that maximizing the log-likelihood (as a function of $\boldsymbol{\beta}$) is equivalent to minimizing the sum-of-squares in (2.2). Therefore, the ordinary least squares estimator

$$(2.5) \quad \hat{\boldsymbol{\beta}} = \left(\sum_{i=1}^n \mathbf{X}_i \mathbf{X}_i' \right)^{-1} \sum_{i=1}^n \mathbf{X}_i Y_i$$

is also the maximum likelihood estimator for $\boldsymbol{\beta}$. The maximum likelihood estimator of the estimator of the variance σ^2 is given by

$$(2.6) \quad \hat{\sigma}^2 = n^{-1} \sum_{i=1}^n (Y_i - \mathbf{X}_i' \hat{\boldsymbol{\beta}})^2$$

which differs from our usual unbiased estimator, by using a divisor of n rather than $n - p$ where p is the number of predictor variables.

Now consider the following setup

$$(2.7) \quad \begin{aligned} &(Y_1, \mathbf{X}_1), \dots, (Y_n, \mathbf{X}_n) \text{ independent sample} \\ &(Y_i | \mathbf{X}_i) \sim \text{Bernoulli}(\pi_i) \end{aligned}$$

which says that $Y_i = 1$ with probability π_i , and $Y_i = 0$ with probability $1 - \pi_i$. Thus the response variable is a dichotomous indicator function which has the properties

$$(2.8) \quad \begin{aligned} E\{Y_i | \mathbf{X}_i\} &= \pi_i \\ V\{Y_i | \mathbf{X}_i\} &= \pi_i(1 - \pi_i) \end{aligned}$$

The natural question to ask is whether we can model the response probability π_i as a function of the predictor variables. One attempt to do this would be to use a *linear probability model* given by

$$(2.9) \quad \pi_i = \mathbf{X}_i' \boldsymbol{\beta}$$

A major criticism with this model is that while the probabilities π_i are, by definition, restricted to fall between zero and one, the estimated regression mean could, in general, take on negative values or values greater than one. In addition, if ordinary least squares is used to fit the linear probability model, then the estimates will not be efficient, and the variances obtained as part of the ordinary least squares regression will not be correct. Fix up's like the ones we have looked at when analyzing data with heteroscedastic variances should be used instead.

Because of the problems associated with the linear probability model, two popular alternatives, *probit models*, and the *logit models* are often used. Both of these models involve nonlinear modeling of the response probability in terms of the regression coefficients $\boldsymbol{\beta}$. Indeed, the regression coefficients take on different interpretations under these models than they do for the linear probability model.

The *probit model* is given by

$$(2.10) \quad \pi_i = \Phi(\mathbf{X}_i' \boldsymbol{\beta}) \quad \text{or} \quad \Phi^{-1}(\pi_i) = \mathbf{X}_i' \boldsymbol{\beta}$$

where Φ is the standard Normal cumulative distribution function. This model is often used in Economics, and is motivated by modeling an underlying latent variable which is Normally

distributed, and modeling its probability of falling into two distinct regimes with probability π_i and $1 - \pi_i$ respectively.

The *logistic model* is given by

$$(2.11) \quad \pi_i = \frac{\exp(\mathbf{X}_i' \boldsymbol{\beta})}{1 + \exp(\mathbf{X}_i' \boldsymbol{\beta})} \quad \text{or} \quad \log\left(\frac{\pi_i}{1 - \pi_i}\right) = \mathbf{X}_i' \boldsymbol{\beta}$$

Notice that the logistic model, says that the *log-odds* can be written as a linear function of the regression coefficients, where the *odds* is defined by

$$(2.12) \quad Odds_i = \frac{\pi_i}{1 - \pi_i}$$

The odds, or odds ratio, transforms the probabilities from the $[0, 1]$ scale to the $[0, \infty)$ scale, and taking the log of the odds further transforms the probabilities to the $(-\infty, \infty)$ scale.

Notice that

$$(2.13) \quad \begin{aligned} \log\left(\frac{\pi_i}{1 - \pi_i}\right) &\rightarrow \infty \quad \text{as} \quad \pi_i \rightarrow 1 \\ \log\left(\frac{\pi_i}{1 - \pi_i}\right) &\rightarrow -\infty \quad \text{as} \quad \pi_i \rightarrow 0 \end{aligned}$$

so that the log odds preserves the ordering of the probabilities.

In the Table 1 we list the values of the Odds and log-Odds for several probabilities

Probability	Odds	Log-Odds
0.0	0.000	$-\infty$
0.1	0.111	-2.197
0.2	0.250	-1.386
0.3	0.429	-0.847
0.4	0.667	-0.405
0.5	1.000	0.000
0.6	1.500	0.405
0.7	2.333	0.847
0.8	4.000	1.386
0.9	9.000	2.197
1.0	∞	∞

The log-odds is symmetric about zero in the same way the probabilities are symmetric about the value 0.5. Note that the definition of Odds used in (2.12) is the inverse of that commonly used in betting where a probability of 0.25 (say) would be associated with *fair odds* of 3 to 1, not 1/3 as given in (2.12).

Given the definition of the model for the observed data given in (2.7) we can write the likelihood as

$$(2.14) \quad L(\boldsymbol{\beta}) = \prod_{i=1}^n \pi_i^{Y_i} (1 - \pi_i)^{1-Y_i}$$

and the log-likelihood can be written as

$$(2.15) \quad \begin{aligned} \log L(\boldsymbol{\beta}) &= \sum_{i=1}^n [Y_i \log \pi_i + (1 - Y_i) \log (1 - \pi_i)] \\ \log L(\boldsymbol{\beta}) &= \sum_{i=1}^n \left[Y_i \log \left(\frac{\pi_i}{1 - \pi_i} \right) + \log (1 - \pi_i) \right] \\ \log L(\boldsymbol{\beta}) &= \sum_{i=1}^n [Y_i \mathbf{X}_i' \boldsymbol{\beta} + \log (1 - \pi_i)] \end{aligned}$$

where we used expression (2.11) to derive the last step. It can be shown that the derivative of the log-likelihood is given by

$$(2.16) \quad \frac{\partial}{\partial \boldsymbol{\beta}} \log L(\boldsymbol{\beta}) = \sum_{i=1}^n \mathbf{X}_i (Y_i - \pi_i)$$

so that setting this expression equal to zero gives the system of nonlinear equations that need to be solved to obtain the maximum likelihood estimates of the regression coefficients $\boldsymbol{\beta}$. The system of equations is nonlinear since the π_i are nonlinear functions of the $\boldsymbol{\beta}$. It can be shown that as the sample size gets large

$$(2.17) \quad \hat{\boldsymbol{\beta}} \text{ is approximately distributed as } N_P \left(\boldsymbol{\beta}, (\mathbf{X}'\mathbf{V}\mathbf{X})^{-1} \right)$$

where

$$(2.18) \quad \begin{aligned} \mathbf{V} &= \text{diag} [\pi_1(1-\pi_1), \pi_2(1-\pi_2), \dots, \pi_n(1-\pi_n)] \\ \mathbf{X} &= \begin{pmatrix} \mathbf{X}'_1 \\ \mathbf{X}'_2 \\ \vdots \\ \mathbf{X}'_n \end{pmatrix} \end{aligned}$$

Expression (2.17) allows us to make inference about the coefficient vector, such as testing some of the coefficients equal to zero, or tests about linear combinations of the coefficients like we have done in the usual linear regression setting. This is analogous to the *testing coefficients method* we discussed in the usual linear regression setting. There is also an analogy to the generalized F -testing procedure we discussed in the usual linear regression setting as well. In order to discuss this, we need to define the notation

$$(2.19) \quad G_p = -2\log L(\hat{\boldsymbol{\beta}})$$

where the p denotes the fact that there are p predictor variables in the full model. Now assume that we want to test whether we can reduce to a model with $k < p$ predictors and define G_k by

expression (2.19) where the coefficient vector is estimated for the reduced model. Then it can be shown that as the sample size gets large

$$(2.20) \quad G_k - G_p = G_{\text{Reduced}} - G_{\text{Full}} \sim \chi^2_{p-k}$$

where χ^2_{p-k} denotes a central chi-square random variable with $p - k$ degrees of freedom.

– This method of testing a full versus reduced model procedure, along with the testing coefficients method made possible by (2.17) provides all the tools we need to make inference about the logistic regression model.

Example 5. We will work an example taken from **Applied Logistic Regression** by David W. Hosmer and Stanley Lemeshow. The data are taken from a study about low birth weight of babies relates to the mother's health practices. There are 189 observations. The variables available for analysis are given below

Table 2. Variables Available for Birth Weight Analysis.

Variable	Description
ID	Identification Code
LOW	Low Birth Weight (0=Birth Weight greater than or equal to 2500g, 1=Birth Weight less than 2500g.)
AGE	Age of Mother in years
LWT	Weight in pounds at the last Menstrual Period
RACE	Race (1=White, 2=Black, 3=Other)
SMOKE	Smoking status During Pregnancy (1=Yes, 0=No)
PTL	History of Premature Labor (0=None, 1=One, etc.)
HT	History of Hypertension (1=Yes, 0=No)
UI	Presence of Uterine Irritability (1=Yes, 0=No)
FTV	Number of Physician Visits During the First Trimester (0=None, 1=One, 2=Two, etc.)
BWT	Birth Weight in grams.

We begin by using Low as the response variable. In addition we created two additional variables RW and RB where

RW = 1 if RACE = White, = 0 Otherwise

RB = 1 if RACE = Black, = 0 Otherwise

The predictors we used for the Full model were Intercept, AGE, LWT, RW, RB, SMOKE, PTL, HT, UI, FTV. The model was fit in SAS using the PROC LOGISTIC procedure with the following code

```
data set1;
  infile 'c:\classnew\class12\birthwt.dat';
  input id low age lwt race smoke ptl ht ui ftv bwt;
  rw = 0;
  rb = 0;
  if (race = 1) then rw = 1;
  if (race = 2) then rb = 1;

proc sort;
  by bwt;

run;

proc logistic order=data;
  model low = age lwt rw rb smoke ptl ht ui ftv;

run;
```

The option order=data tells SAS to use the sort order for the levels of the response variable. Since Low= (0, 1), SAS models Pr(Low=1), i.e., the highest value of the response variable. The output from the running of this program is given next.

The LOGISTIC Procedure

Model Information

Data Set	WORK.SET1
Response Variable	low
Number of Response Levels	2
Model	binary logit
Optimization Technique	Fisher's scoring

Response Profile

Ordered Value	low	Total Frequency
1	1	59
2	0	130

Testing Global Null Hypothesis: BETA=0

Intercept
Intercept and

Intercept
and

Criterion	Only	Covariates
AIC	236.672	221.285
SC	239.914	253.702
-2 Log L	234.672	201.285

Analysis of Maximum Likelihood Estimates

Parameter	DF	Estimate	Standard Error	Wald Chi-Square	Pr > ChiSq
Intercept	1	1.3611	1.1046	1.5184	0.2179
age	1	-0.0295	0.0370	0.6367	0.4249
lwt	1	-0.0154	0.00692	4.9691	0.0258
rw	1	-0.8805	0.4408	3.9902	0.0458
rb	1	0.3918	0.5376	0.5310	0.4662
smoke	1	0.9388	0.4022	5.4501	0.0196
ptl	1	0.5433	0.3454	2.4745	0.1157
ht	1	1.8633	0.6975	7.1356	0.0076
ui	1	0.7676	0.4593	2.7931	0.0947
ftv	1	0.0653	0.1724	0.1435	0.7048

Association of Predicted Probabilities and Observed Responses

Percent Concordant	74.4	Somers' D	0.492
Percent Discordant	25.3	Gamma	0.493
Percent Tied	0.3	Tau-a	0.212
Pairs	7670	c	0.746+

Concordant/Discordant os interpreted as follows. The predicted score for unit k is $\hat{\pi}_k$.

A concordant pair (k_1, k_2) is a pair of observations with (0,1) response values. If $\hat{\pi}_{k_1} < \hat{\pi}_{k_2}$, the pair is concordant; if not, it is discordant.

From the output we obtain the log-likelihood ratio for the full model, as well as the logistic regression coefficient estimates and their standard errors. To illustrate the technique of fitting a reduced model, we tried fitting the reduced model which only contained the predictor variables Intercept, LWT, RW, SMOKE, HT, UI. The additional SAS code to run this is given by

```
proc logistic order=data;
  model low = lwt rw smoke ht ui;
run;
```

and the output is

Model Information

Data Set	WORK.SET1
Response Variable	low
Number of Response Levels	2
Model	binary logit
Optimization Technique	Fisher's scoring

Number of Observations Read	189
Number of Observations Used	189

Response Profile

Ordered Value	low	Total Frequency
1	1	59
2	0	130

Testing Global Null Hypothesis: BETA=0

Test	Chi-Square	DF	Pr > ChiSq
Likelihood Ratio	29.9064	5	<.0001
Score	27.8158	5	<.0001
Wald	23.4610	5	0.0003

Analysis of Maximum Likelihood Estimates

Parameter	DF	Estimate	Standard Error	Wald Chi-Square	Pr > ChiSq
Intercept	1	0.9049	0.8377	1.1667	0.2801
lwt	1	-0.0153	0.00646	5.6077	0.0179
rw	1	-1.0648	0.3881	7.5266	0.0061
smoke	1	1.0912	0.3866	7.9656	0.0048
ht	1	1.8533	0.6882	7.2510	0.0071
ui	1	0.8926	0.4495	3.9426	0.0471

Odds Ratio Estimates

Effect	Point Estimate	95% Wald Confidence Limits
lwt	0.985	0.972 0.997
rw	0.345	0.161 0.738
smoke	2.978	1.396 6.353
ht	6.381	1.656 24.587
ui	2.441	1.012 5.892

Association of Predicted Probabilities and Observed Responses

Percent Concordant	73.2	Somers' D	0.469
Percent Discordant	26.3	Gamma	0.471
Percent Tied	0.5	Tau-a	0.202
Pairs	7670	c	0.734

We can construct the reduced model test as follows

$$G_{10} = 201.285 \text{ with } 9 \text{ df}$$

$$G_6 = 204.766 \text{ with } 5 \text{ df}$$

$$G_6 - G_{10} = 3.481 \text{ with } 9-5 = 4 \text{ df}$$

which is not significant at the 0.01 level since $\chi^2_{4,.01} = 13.28$.

3. Odds Ratios in Logistic Models

Consider independent variable k in the model. Imagine changing this variable by 1 unit while holding all others constant.

The log of the ratio of the odds of having the characteristic versus not having the characteristic, given data value $x_k + 1$ for unit i , to odds of having the characteristic versus not having the characteristic, given data value x_k , is equal to

$$\begin{aligned} \log \left\{ \frac{\pi_i(x_k + 1) / [1 - \pi_i(x_k + 1)]}{\pi_i(x_k) / [1 - \pi_i(x_k)]} \right\} &= \\ \log \left\{ \pi_i(x_k + 1) / [1 - \pi_i(x_k + 1)] \right\} &- \\ \log \left\{ \pi_i(x_k) / [1 - \pi_i(x_k)] \right\} &= \\ = (x_k + 1) \beta_k - x_k \beta_k &= \\ = \beta_k \end{aligned}$$

$\exp(\beta_k)$ is odds ratio for two cases that are 1-unit apart on the k^{th} independent variable with all other explanatory variables held constant.

If x_{ik} is an indicator for sex, e.g., $x_{ik} = 1$ if male and 0 if female, then $\exp(\beta_k)$ is ratio of the odds for a male of having the characteristic to the odds for a female of having the characteristic.

If you have a standard set of reference categories, this makes interpretation of odds ratios easier to remember. For example, if non-Hispanic white is the reference category for race-ethnicity, then the OR's measure how much more or less at risk a group is compared to non-Hispanic whites.

Example: Odds Ratio in Main Effects Model

- One categorical factor: sex; predict whether someone has a disease
- Probability of male having disease is 0.2; odds = $0.2/0.8 = 0.25$
- Probability of female having disease is 0.1; odds = $0.1/0.9 = 0.11$
- **Relative risk** of males compared to females is $p_M / p_F = 0.2/0.1 = 2$
- Ratio of the odds (**Odds ratio**) of male having disease compared to female having disease is $0.25/0.11 = 2.25$

Colloquial interpretation:

“Males are 2.25 times more likely to have disease than females”

These words really go better with the relative risk.

If probabilities of having the disease are very small, then the colloquial interpretation of the odds ratio is very close to the truth since $1 - p \doteq 1$. For example, if the probability of a male having the disease is 0.02 and a female 0.01, then the odds ratio is

$$\frac{0.02/0.98}{0.01/0.99} = 2.02$$

which is close to $0.02/0.01 = 2.00$. We will see more of this in Notes for Class #13 in the context of 2x2 tables.

Two factor model with no interactions

- Model for the logit of the proportion, π_{ij} , in level i of the first factor and level j of the second factor contains an intercept and main effects:

$$\text{logit}(\pi_{ij}) = \log\left(\frac{\pi_{ij}}{1 - \pi_{ij}}\right) = \mu + \alpha_i + \beta_j.$$

- The log-odds ratio for the (1,1) cell vs. the (1,2) cell is

$$\begin{aligned}\log\left[\frac{\pi_{11}/(1 - \pi_{11})}{\pi_{12}/(1 - \pi_{12})}\right] &= (\mu + \alpha_1 + \beta_1) - (\mu + \alpha_1 + \beta_2) \\ &= \beta_1 - \beta_2\end{aligned}$$

- Corresponding odds-ratio is $\exp(\beta_1 - \beta_2)$
- If β_2 is for the reference level, then the estimated odds of having characteristic for someone in cell (1,1) vs. someone in the reference cell (1,2) is $\exp(\hat{\beta}_1)$ since the solution for β_2 is set to 0.
- If there were other independent variables in the model, they would subtract in the log-odds calculation above as long as we keep their values the same in the numerator and denominator odds.

Confidence Intervals for Odds Ratios

A two-sided $(1-\alpha)$ -level confidence interval for an odds ratio can be found by putting a confidence interval on the parameter β_k (or a linear combination of parameters) and then transforming to the odds ratio scale. For the case of a single β_k , the lower and upper confidence bounds are

$$\text{Lower bound: } \exp\left[\hat{\beta}_k - t_{1-\alpha/2, DF} \sqrt{\text{var}(\hat{\beta}_k)}\right]$$

$$\text{Upper bound: } \exp\left[\hat{\beta}_k + t_{1-\alpha/2, DF} \sqrt{\text{var}(\hat{\beta}_k)}\right]$$

where $\text{var}(\hat{\beta}_k)$ is the estimated variance of $\hat{\beta}_k$ and $t_{1-\alpha/2, DF}$ the $(1-\alpha/2)$ percentile from the t distribution with DF degrees of freedom.

```
*****;
* FILE:      C:\JPSM\SURV 616\class12\logistic.sas      ;
* TOPIC:     Illustrate use of proc logistic in SAS using low birthweight;
*            dataset from Hosmer and Lemeshow (2000)    ;
*            www.umass.edu/statdata/statdata/stat-logistic.html ;
* DATE:      04/13/09                                  ;
*****;

libname in 'C:\JPSM\SURV 616\class12';

title 'Occurrence of low birthweight';

proc logistic data = in.lowbwt;
    class race (descending)
           smoke (descending)
           ht (descending)
           ui (descending);
    model low (event = '1') = age lwt ptl ftv
                                race smoke ht ui
                                ;
    output out = preds predicted = pi_hat;
run;

proc contents data = preds;
run;

data preds;
    set preds;
    l_odds = log( pi_hat / (1-pi_hat) );
run;

ods graphics on;
```

```
ods html;

symbol1 value=dot;
proc gplot data = preds;
    plot pi_hat * lwt = low;
    plot l_odds * lwt = low;
run;

ods graphics off;
```

The LOGISTIC Procedure

Model Information

Data Set	IN.LOWBWT
Response Variable	LOW
Number of Response Levels	2
Model	binary logit
Optimization Technique	Fisher's scoring

Number of Observations Read	189
Number of Observations Used	189

Response Profile

Ordered Value	LOW	Total Frequency
1	0	130
2	1	59

Probability modeled is LOW=1.

Class Level Information

Design Class	Value	Variables
RACE	3	1 0
	2	0 1
	1	-1 -1
SMOKE	1	1
	0	-1
HT	1	1
	0	-1
UI	1	1
	0	-1

Model Convergence Status

Convergence criterion (GCONV=1E-8) satisfied.

Model Fit Statistics

Criterion	Intercept Intercept Only and Covariates	
	Intercept Only	Intercept and Covariates
AIC	236.672	221.285
SC	239.914	253.702
-2 Log L	234.672	201.285

The LOGISTIC Procedure

Testing Global Null Hypothesis: BETA=0

Test	Chi-Square	DF	Pr > ChiSq
Likelihood Ratio	33.3872	9	0.0001
Score	30.9586	9	0.0003
Wald	25.7025	9	0.0023

Analysis of Maximum Likelihood Estimates

Parameter	DF	Standard		Wald	
		Estimate	Error	Chi-Square	Pr > ChiSq
Intercept	1	2.9831	1.2478	5.7158	0.0168
AGE	1	-0.0295	0.0370	0.6367	0.4249
LWT	1	-0.0154	0.00692	4.9691	0.0258
PTL	1	0.5433	0.3454	2.4745	0.1157
FTV	1	0.0653	0.1724	0.1435	0.7048
RACE	3	0.1629	0.2766	0.3469	0.5559
RACE	2	0.5547	0.3232	2.9458	0.0861
SMOKE	1	0.4694	0.2011	5.4501	0.0196
HT	1	0.9317	0.3488	7.1356	0.0076
UI	1	0.3838	0.2297	2.7931	0.0947

Odds Ratio Estimates

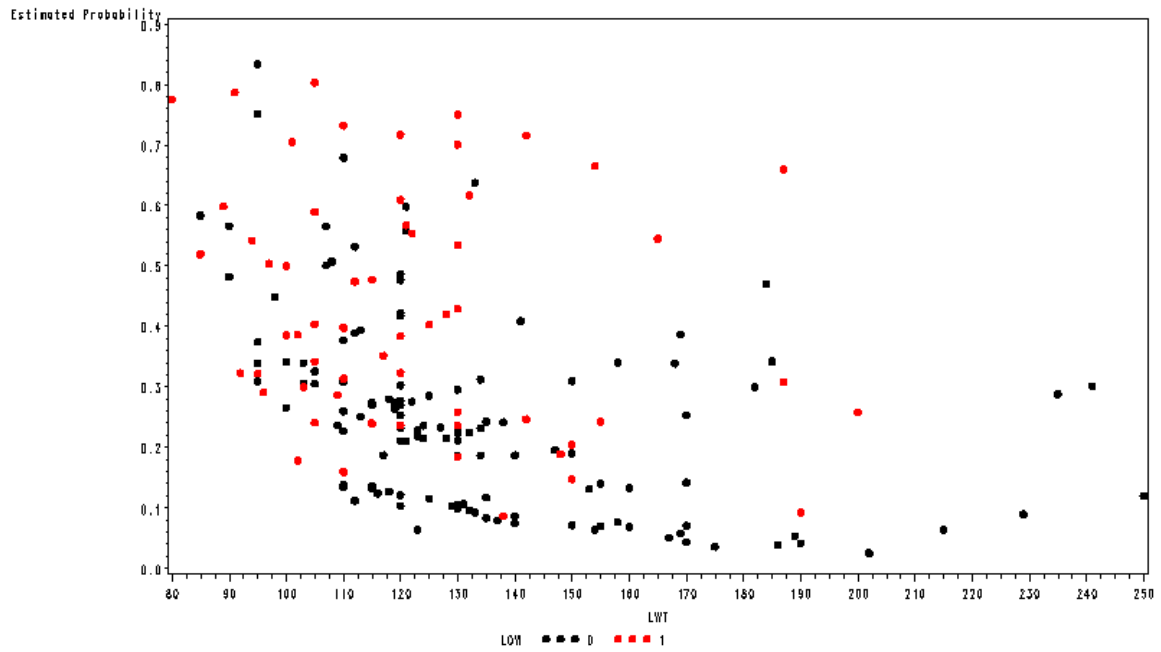
Effect		Point Estimate	95% Wald Confidence Limits	
AGE		0.971	0.903	1.044
LWT		0.985	0.971	0.998
PTL		1.722	0.875	3.388
FTV		1.067	0.761	1.497
RACE	3 vs 1	2.412	1.017	5.723
RACE	2 vs 1	3.569	1.270	10.033
SMOKE	1 vs 0	2.557	1.163	5.624
HT	1 vs 0	6.445	1.642	25.291
UI	1 vs 0	2.155	0.876	5.301

The LOGISTIC Procedure

Association of Predicted Probabilities and Observed Responses

Percent Concordant	74.4	Somers' D	0.492
Percent Discordant	25.3	Gamma	0.493
Percent Tied	0.3	Tau-a	0.212
Pairs	7670	c	0.746

Occurrence of low birthweight



Occurrence of low birthweight

